

Ashwin Sateesh Kumar

Palo Alto, CA | 925-445-6494 | ashwinsateeshkumar@gmail.com | [linkedin.com/in/ashwins24](https://www.linkedin.com/in/ashwins24) | github.com/ashwin-sateesh

EXPERIENCE

AI Software Engineer

Dec. 2024 – Present

Citibank

Palo Alto, CA

- Reduced false positives by 30% and improved detection accuracy by 25%, by building a dependency-aware deep code analyzer with source-to-sink reasoning into a ReAct-based multi-agent vulnerability detection system
- Cut manual penetration testing lead times by over 50% by integrating a report generation agent that delivers structured vulnerability analysis, exploit simulation, and remediation guidance into the ReAct pipeline
- Productionized a LangGraph-based multi-agent security platform with threat modeling, architecture diagram, and cloud log analysis workflows exposed via custom MCP server, deployed to 8,000+ analysts
- Ensured release quality by integrating a LLM-powered E2E testing framework built on Playwright into CI/CD pipeline
- Architected a microservices-based RAG system with BM25 + cross-encoder reranking, RAGAS evaluation, prompt caching, and async chat history, improving retrieval precision by 25% and cutting redundant vector lookups by 40%

Machine Learning Engineer

Jan. 2024 – Dec. 2024

Abecedarian

Boston, MA

- Led the design of a multimodal Yoga Assistant by fine-tuning GPT-3.5 with parallel queue-based continual learning and automated web scraping for real-time updates, integrating Chainlit UI and perplexity-based benchmarking
- Enhanced image generation by fine-tuning Stable Diffusion and distilling into lightweight Latent Diffusion Model via U-Net pruning for yoga posture visualization, achieving 2x faster inference while preserving CLIP alignment

Research Assistant

July 2023 – Nov. 2023

Northeastern University

Boston, MA

- Uncovered distinct visual and linguistic patterns in social media by training a custom disentangled multimodal β -VAE(U-Net for images and BERT for text) and analyzing learned interpretable latent factors from 400k posts
- Achieved fine-grained control over multimodal generation by manipulating disentangled latent factors; deployed the system on GCP with bfloat16 quantization, reducing inference latency by 30%

Machine Learning Engineer Intern, R&D

June 2022 – Dec. 2022

Signify (Phillips Lighting)

Boston, MA

- Engineered a user re-identification system for personalized home automation, achieving 0.95 mean average precision (mAP) in identity matching across multi-camera views using an omni-scale feature learning architecture
- Developed a multispectral vision pipeline using infrared and RGB imagery with Faster R-CNN ROI localization, and U-Net segmentation for plant anomaly detection, driving real-time closed-loop lighting optimizations

Trainee Software Engineer

July 2019 – Nov. 2020

KPIT Technologies Ltd.

Bengaluru, India

- Developed an autonomous driving perception system using U-Net traffic scene segmentation with 0.89 Intersection over Union (IoU) and vision-radar sensor fusion with kinematic features for real-time object detection

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), R

Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, Hugging Face, FastAPI, Flask

Developer Tools: Git, Bash, Docker, GCP (Vertex AI, Cloud Storage), AWS (S3, SageMaker), Red Hat OpenShift, Tekton

Libraries: NumPy, pandas, SciPy, OpenCV, NLTK, XGBoost, FAISS

PROJECTS

HealthBot | Python, PyTorch, Flask, HTML, Docker

Sep. 2023 – Dec. 2023

- Developed a chatbot generating medically relevant responses (0.75 semantic similarity) by fine-tuning FLAN-T5 with LoRA and RLHF (PPO), integrating BERT named entity recognition and disease classification (0.96 F1)

EDUCATION

Northeastern University

Boston, MA

Master of Science in Data Science (GPA: 3.8/4.0)

Sep. 2021 – Dec. 2023

PES University

Bengaluru, India

Bachelor of Engineering in Electronics and Communications

Aug. 2015 – May 2019